

HUJI Applicative Research Report — Exact & Social Sciences — Week 36, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 36, 2026 · Generated August 31, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights two groundbreaking research projects from Hebrew University with significant commercial potential. One study introduces an advanced methodology for quantifying subtle feature detectability under challenging conditions, offering immediate applications in automated quality control and forensic analysis. The second research delves into how individuals internalize emotional patterns, paving the way for more sophisticated AI-driven social intelligence and behavioral modeling in human-computer interaction and digital mental health applications. Both underscore the university's expertise in developing intelligent systems for real-world impact.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Naomi Kaplan-Damary — Software/AI, Other
- Maya Tamir — Software/AI, Neuroscience, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Evaluating the detectability of randomly acquired characteristics under crime scene-like conditions. 14/50

"Novel AI for detecting subtle patterns promises revolution in manufacturing quality control and forensics."

Main researcher: Naomi Kaplan-Damary · naomi.kaplan@mail.huji.ac.il.

Department of Statistics and Data Science, The Hebrew University of Jerusalem, Israel.

Published · 2026-05

ABOUT THIS RESEARCH

The study investigates how easily unique, random marks on shoe soles can be detected and collected from various surfaces under realistic crime scene conditions. Researchers used statistical modeling to determine the probability of identifying these specific characteristics across different materials and shoe types.

COMMERCIAL ANGLE

This research methodology for quantifying feature detectability under noise could be adapted into computer vision software for automated forensic pattern recognition or quality control in precision manufacturing.

COMMERCIALISATION METRICS

Novelty

3/10

The research is an empirical validation study of existing forensic principles rather than a new methodology or technology. It provides valuable statistical data on detection rates, but it does not introduce a novel scientific concept or a groundbreaking technical innovation.

Commercial Potential

3/10

The study is fundamental forensic research focused on validating existing investigative methodologies rather than developing a new, scalable product, service, or platform technology. There is no clear path to intellectual property, a proprietary device, or a commercialized software solution presented in the work.

Market Size

2/10

The study is a specialized forensic validation experiment rather than a product or technology with scalable market potential. The addressable market is limited to niche law enforcement training or specific forensic tool validation, offering very low commercial value.

Tech Readiness

4/10

The study is currently in the fundamental research and validation stage, focusing on characterizing detection probabilities under controlled experimental conditions rather than deploying a commercial tool. While it provides necessary empirical data for forensic protocols, it lacks the integration into a commercial product or a field-ready diagnostic platform.

IP Strength

2/10

The paper is a forensic methodology study focused on observational data and statistical analysis rather than the development of a novel device, chemical reagent, or proprietary algorithm. There is no evidence of patentable technical innovation or proprietary IP that would provide a competitive moat.

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2. Learning affect norms: Implications for predictions, experiences, and social judgments.

12/50

"AI learns human emotional norms, paving the way for empathetic digital health and social intelligence tools."

Main researcher: Maya Tamir

The Hebrew University of Jerusalem, Department of Psychology.
Published in Journal of experimental psychology. General · 2026-05

ABOUT THIS RESEARCH

The study demonstrates that individuals observe and internalize emotional patterns within a group, creating 'affect norms' that influence their own feelings and social judgments. These learned norms serve as benchmarks that cause people to predict group behavior and judge those who deviate from the norm more negatively.

COMMERCIAL ANGLE

This research can inform the development of advanced AI-driven social intelligence tools and behavioral modeling software for digital mental health, social media moderation, and human-computer interaction.

COMMERCIALISATION METRICS

Novelty

4/10

The research provides incremental psychological insight by extending dyadic emotion studies to group-level norm formation, but it does not introduce a novel mechanism or transformative framework. It is a theoretical refinement of existing social cognition principles rather than a groundbreaking shift in the field.

Commercial Potential

3/10

The paper is a fundamental contribution to social psychology regarding behavioral mechanisms rather than a tangible technology or therapeutic intervention. While it offers theoretical insights that could inform social AI or UX design, it lacks a direct path to a proprietary product, clinical application, or licensable deep-tech platform.

Market Size

2/10

The paper is a fundamental study in social psychology exploring cognitive mechanisms rather than a technological or therapeutic intervention with a clear commercial application. There is no defined product, service, or industrial use case that translates these theoretical findings into a scalable, addressable market.

Tech Readiness

2/10

This is fundamental basic research in social psychology conducted in a controlled lab setting via human subject studies. It lacks any applied technological component, prototype, or direct path to a commercialized product or clinical intervention.

IP Strength

1/10

This is a social psychology research paper exploring behavioral phenomena and human cognitive processes, which are natural laws and cannot be patented. There is no proprietary technology, algorithm, or physical invention described that offers industrial or clinical defensibility.

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